

Software Fiável

Assignment 1

Hoare Logic

Mestrado em Engenharia de Informática
Mestrado em Segurança Informática
Mestrado em Informática
Faculdade de Ciências da Universidade de Lisboa

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The purpose of this assignment is to help you develop your understanding of Hoare logic and your ability to use it to reason about program correctness. As you work through the three questions, ensure that you understand and can justify everything you include in your solutions, as grading will also be based on an oral examination assessing your individual understanding of the work you submit.

1 Max of three values (0.5 points)

Consider the program P below, where program variables a, b, c, max (a.k.a. storage cells) store integers.

```
if a >= b && a >= c then
  max := a;
else
  if b >= a && b >= c then
    max := b
  else
    max := c
```

1. Present a proof that the triple below is valid in the form of a derivation tree or in the form of an annotated program (your choice).

$$\{true\} P \{max \geq a \wedge max \geq b \wedge max \geq c\}$$

2. The triple above, although valid, is imprecise. Write a valid triple that serves as a better specification of the program.

2 Nines (0.75 points)

Consider the program *Nines* below, over integer variables.

```
i := 1;  
r := 9;  
while i < n do  
  r := r * 10 + 9;  
  i := i + 1;
```

1. Prove the validity of the triple below.

$$\{n \geq 1\} P \{r = 10^n - 1\}$$

Present your proof in the form of a derivation tree or in the form of an annotated program (your choice). Clearly identify the invariant. Be sure to thoroughly justify all implications.

2. If we consider a stronger precondition, say $\{n > 1\}$, is the triple still valid? Justify your answer.
3. Prove that *Nines* terminates if the input provided in n is positive.

3 Soundness (0.25 points)

The assignment axiom looks backward to almost everyone the first time they see it. If it still seems backward to you, it may help to think a little about alternative “forward” axiom. Here is a seemingly natural one:

$$\{\text{true}\} x := e \{x = e\}$$

Explain what is wrong with using this triple as an axiom. Hint: what happens if x occurs in e ?

Delivering your assignment

- Deadline: October 13, 2026, 23:59.
- Your assignment is composed of *one* PDF file. Handwritten assignments are OK provided they are clean, carefully written, and with a clear handwriting. Upload it at the course’s web site before the deadline.

Oral examination

The oral examination will take place shortly after the assignment submission deadline in room 6.3.16. This schedule[↗] is available on the course’s Moodle page for you to book a 10-minute slot. If none of the available slots suit you, you can ask for an alternative.